HSAN ELLOUZE

Software Engineering Student | Enterprise AI & Semantic Search Systems
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Seeking a 6-month end-of-studies internship (PFE) starting February 2027 | Open to relocation (Walldorf)

Summary

Final-year Software Engineering student (INSAT, expected graduation September 2027) with production experience building enterprise AI systems that give grounded answers over curated source documents: bilingual RAG chatbots, LLM-backed contract analysis, and multi-agent diagnosis tooling. Hands-on across document ingestion, structure-aware chunking, embeddings and semantic search, LLM orchestration (LangChain/LangGraph) and multi-provider evaluation (OpenAI, Claude, Gemini). Grounded in the backend and fullstack engineering (FastAPI, Express.js, React/Next.js, Docker/CI-CD) needed to ship AI features inside an enterprise platform.

Experience

AI Engineer Intern – Applicare, Canada Jun 2026 – Aug 2026
- Architected AskSahera, a bilingual (FR/EN) production RAG chatbot giving **grounded answers** over ∼100 curated source documents
- Developed a dual-source ingestion pipeline (PubMed XML + GROBID PDF parsing) with IMRAD-aware, **structure-aware chunking** and a sliding-window fallback
- Built the RAG backend in Express.js with Qdrant vector storage and OpenAI embeddings for **embedding-based retrieval**, using Claude for generation; integrated Cloudflare R2 for storage

Mazdev – Tunis
Software Engineer Intern Jul 2025 – Sep 2025
- Created a RAG engine (Pinecone) reaching 94% **semantic search** accuracy across a large document set; optimized embeddings to cut latency 45%
- Engineered a production chatbot platform (YAML-configurable) used by non-technical users to create 20+ bot variants
- Established an automated testing framework, cutting the QA cycle 50%

Fullstack Developer Mar 2025 – Jun 2025
- Architected an AI-powered legal contract review platform for 40+ enterprise clients, cutting analysis time 60%
- Delivered a React/Next.js + FastAPI RESTful API system processing 500+ contract reviews per month at production scale
- Implemented a Docker CI/CD pipeline, cutting infra setup time 75%

Frontend Developer Intern – Steros, Sfax Jul 2024 – Aug 2024
- Built a responsive showcase website and interactive UI components, with a PHP/MySQL backend

Projects

KubeMind: AI-Driven Kubernetes Observability Platform (3-person capstone team) GitHub
- Designed a 5-layer **agentic** software architecture (ingestion, AI agents, correlation/storage, presentation) for automated incident diagnosis
- Implemented 3 specialized LangChain/LangGraph agents (logs, metrics, events) orchestrated via a LangGraph StateGraph workflow, feeding a correlation agent
- Developed a FastAPI backend and Next.js dashboard on a real-time ingestion pipeline (Prometheus, Loki, Kafka, MongoDB)
- Benchmarked output quality across 3 LLM providers (OpenAI, Gemini, Ollama) across 3 controlled incident scenarios

Multi-Tenant AI Chatbot Widget GitHub
- Designed a reusable, embeddable AI chatbot component for multi-tenant deployment across client websites via a single `<script>` tag
- Enforced Shadow DOM isolation and strict domain validation for secure multi-tenant use
- Integrated the backend with Hono, Supabase, and Upstash Redis; deployed on Railway and Cloudflare Pages

Education

INSAT – National Institute of Applied Sciences and Technology Sep 2022 – Sep 2027 (expected)
Engineering Degree in Software Engineering (EQF Level 7 / Master's equivalent)

Skills

AI/ML: LLM Integration, RAG, LangChain, LangGraph, Vector Databases (Qdrant, Pinecone), Embeddings, Prompt Engineering
Languages: Python, JavaScript, TypeScript, Java, C#, PHP, SQL
Backend: FastAPI (Python), NestJS (JavaScript), Express.js (JavaScript), Hono (TypeScript), ASP.NET Core (C#)
Data & Observability: MongoDB, MySQL, Supabase, Redis (Upstash), Prometheus, Loki, Kafka
DevOps & Cloud: Docker, Kubernetes, GitHub Actions, CI/CD, Vercel, Railway, Cloudflare (Pages, R2)
Frontend: React, Next.js, Vue
Tools: Git, Linux

Activities

Event Mission Leader – Junior Enterprise INSAT 2024 – 2025
- Supervised a ∼40-member team over the 2024–2025 mandate, leading the organizing committee for 2 flagship events, including Forum INSAT Entreprise (1,500+ attendees, 40+ companies)

Operations Manager – WinterCup 7.0 2025
- Owned end-to-end logistics and event planning for WinterCup 7.0, an overnight coding contest with ∼300 participants, coordinating the on-site operations staff under time pressure